Optimizing Online Interval Scheduling Systems with Data-Driven Analytics for Enhanced Resource Utilization through Online Algorithms

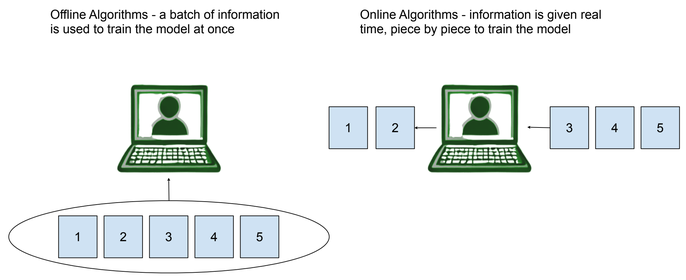
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Aspiring Scientists Summer Internship Program

Background

Modern scheduling mechanisms fail to mold to the variety of constraints posed by real world time management. This ultimately causes a surplus of wasted time for both the client's and administration's schedules. Offline algorithms, which are commonly used in this field, are algorithms that are trained on an input where the entire dataset is fed into the algorithm immediately. Conversely, online algorithms receive input incrementally and must make decisions without full data, learning more and more over time.



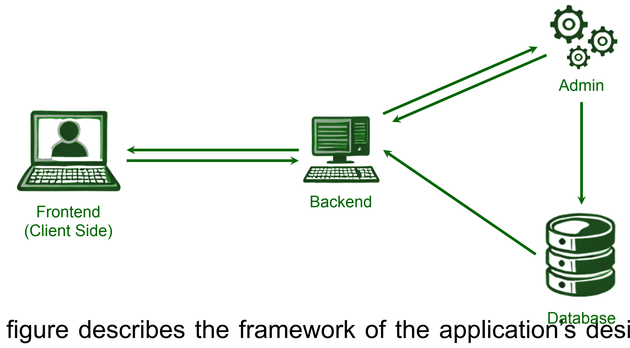
Materials and Methods

A Django framework was used to develop an online interval scheduling system with a Python backend and an HTML frontend, supported by an SQL database. Our algorithm employed a "follow-the-leader" approach, selecting from three sub-algorithms prioritizing room popularity, density, and current reservations. User data was gathered via a website and processed to optimize room allocation. All coding was done using standard computer equipment.

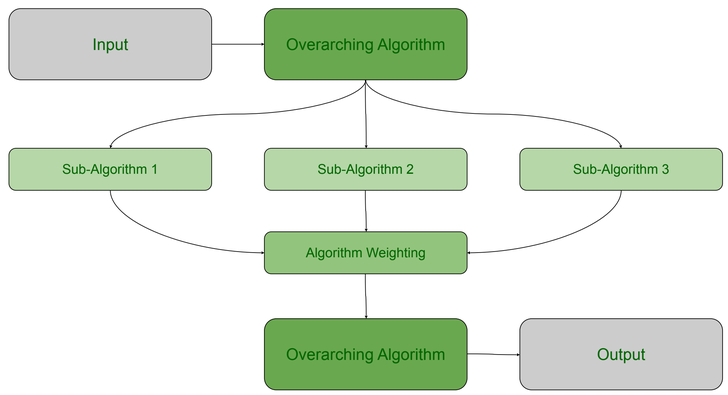


Results

This figure illustrates the basics of the interval scheduling this project pioneers. On day 1, the first request is to find a 2 hour slot between 12pm and 6pm. With no prior data the algorithm randomly selects a 2 hour slot to assign to the client. The second client requests a slot from 4pm-10pm. The algorithm randomly returns either 6pm-8pm or 8pm-10pm. 4pm-6pm is not considered as the first client shows that slot is more popular than the others. On Day 2, using data from the previous day, 8pm-10pm is the least popular so it's returned. This process is continued over multiple days increasing the precision of the online algorithm.



This figure describes the framework of the application's design. The design is centered around the backend controlled by a python based Django System. The information of the rooms can be displayed on the client side and requests can be received from the client side all through a website. Django operates with SQLite, a common language to query data. Django interfaces SQLite into python, allowing programmers to query the database using python. The Admin side has the ability to edit the database and interact with the backend.



The second flowchart characterizes the interaction between the frontend and the backend. The algorithm receives client requests from the front end and runs it through 3 sub algorithms. The overarching algorithm then weighs the conclusions from each sub algorithm and returns the best interval.

Conclusions and Future Work

This application established the scaffolding to build upon the complexity of the algorithm created. The interval scheduling created examines the popularity of specific slots from the entire database. With more data, future work can implement a rolling horizon approach basing the algorithm off only recent data. Future Work can also create more fields in the collected data Such as weather patterns for the specific day, allowing the algorithm to narrow its scope of data and return a increasingly precise interval. Adding such capabilities require the algorithm to run through more data. Future work will center on feeding the algorithm and slowly adding more fields/sub-algorithms for the algorithm to consider.

Major Citations

Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). Introduction to algorithms (4th ed.). The MIT Press. https://doi.org/10.1080/10509585.2015.1092083

Hoi, S., Lu, J., & Zhao, P. (2018, October). Online learning: A comprehensive survey. arXiv. Retrieved July 31, 2024, from https://arxiv.org/pdf/1802.02871

Li, J. (2024, February 17). Online learning, the hedge algorithm, and solving LPs [Unpublished manuscript]. https://www.cs.cmu.edu/~15850/notes/jason-lecture17.pdf

Acknowledgements

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